# Generalized Approachability Results for Stochastic Games with a Single Reachable State

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**Abstract.** The theory of approachability concerns repeated games with vector-valued reward. A set in the reward space is called *approachable* if the decision maker can ensure that his long term average reward vector is in the set, no matter what the adversary may do. Approachability theory was introduced for repeated matrix games by Blackwell (1956), where geometric conditions and related approaching policies were given. The theory was later extended to stochastic (Markov) games under the assumption that some fixed state is recurrent under all stationary strategies. In this work we generalize the latter to games with a single reachable state structure. In these games there exists a single state to which a return from any state may be guaranteed by at least one of the players.

**Keywords:** Stochastic games, vector valued reward, approachability theory.

## 1 Introduction

The theory of approachability for repeated matrix games was introduced by Blackwell in his seminal paper [Bla56]. Approachability theory deals with the case of a vector-valued reward function and average reward criteria. This theory characterizes those sets for which a player can guarantee that his long-term average reward will reside in. A set is *approachable* if the player can make sure that no matter what his opponent does, the average reward will eventually reside in this set.

Approachability theory was first extended to stochastic games in [SS93]. The results there rely on the assumption that a particular reference state is recurrent under any pair of strategies. It is further assumed in [SS93], for the case of an infinite state space, that the return to the reference

state is bounded in expectation. Our results can be considered an extension of [SS93], as we rely on the assumption that a single state *can* be returned to in bounded time. This assumption replaces the recurrence assumption as it does not require that the reference state is recurrent under *all* strategies. [SS93] suggested a piecewise stationary approaching policy that was based on switching between stationary strategies (each optimal for an appropriate projected-reward zero sum game) at returns to the reference state; as was observed there, this guarantees that the long-term average value of the projected game is attained in expected value over each renewal period. The approaching strategy suggested by our scheme is somewhat more complicated, as stopping times with this property need not exist, and one needs to rely on a gradual increase of the time between policy switches. By assuming that there is a state which is reachable by both players, or more generally that the average reward projected game has the same value from all states, we retain the approachability/excludability duality for convex sets, and provide necessary and sufficient conditions for a convex set to be approachable. Finally we provide the needed machinery for extending our results to the infinite state space, similarly to [SS93].

A different type of approachability results for stochastic games was recently presented in [Mil00]. The basic assumption there is a finite cardinality of the state space. The result is based on a recent remarkable theorem from [Mil00], this theorem states that the value of the $n$ steps game converges to the value of the infinite horizon average reward game, uniformly (in the parameters of the game) and with polynomial rate. While the results of [Mil00] rely on the finiteness of the states space, we do not rely on it explicitly, but rather require proper stability assumptions. For finite state space our sufficient conditions for approachability may be considered a specialization of the results of [Mil00]. The reachability assumption allows us to characterize approachable convex sets, similarly to [Bla56] and [SS93], as sets whose intersection with all the sets in a certain class is not empty. This characterization is relatively easy to check and most useful when applying approachability theory, as demonstrated in [MS00]. By assuming that there is a state which is reachable by both players we retain the excludability/approachability duality for convex sets, similarly to [Bla56] and [SS93].

There are two basic ideas in our work that allow us to generalize the results of [SS93]. The first idea is that the main approachability result there still holds if the renewal time is replaced by any finite stopping time of bounded variance. Second, we make the observation that if all expansions of a set are approachable then so is the set itself. This observation suggests that approachability is essentially a topological property of a set and a game. We use both ideas to relax the stability requirements.

The single state reachability assumption may be more natural than the stronger single-state recurrence conditions of [SS93] in various applications. For example, in the queuing problem considered there it was necessary to impose non-idling conditions on the server in order to assure that the zero state is inevitably reached. This condition, which is non required under the present

framework, seems reasonable in many situations. A common situation in service and communications systems is that a "reset" operation is possible, but may be costly so that it is not used under some optimal policies. In a competitive queue example this operation represents a situation in which the queue discards all present messages and starts afresh. We finally note that the single state reachability assumption is perhaps the weakest explicit one that leads to a state-invariant value for all possible reward functions, as required to retain our results for convex sets

Approachability theory has found various uses in game theory and related applications; see, e.g., [FL95, HMC98, Leh98, Spi99, Rus99] and the recent special issue [GAE99]. Approachability in stochastic games was demonstrated in [SS93] for analysis of competitive queues and recently in [MS00] for defining regret minimization for stochastic games. A notable advantage of Blackwell's original approach, which is true for our approach as well, is that no strategic assumptions are made concerning the opponent's strategy. Furthermore, the distance of the average reward from the set decreases uniformly over the opponent's strategies.

The paper is organized as follows. Section 2 presents the stochastic game model and recalls the basic definitions related to approachability theory. In Section 3 we provide an abstract approachability result for stochastic games. Section 4 considers the case of convex sets and provides a necessary and sufficient condition for the approachability of a set. Section 5 presents the required additional assumptions needed for the case of infinite number of states. In Section 6 we provide an illustrative example in which the assumptions of [SS93] do not apply.

## 2 The General Model

We consider a two-person stochastic game with a finite action space per state, which is played ad-infinitum. We refer to the players as P1 (the regret-minimizing player) and P2. The game is defined by the five-tuple $(\mathcal{S}, \mathcal{A}, \mathcal{B}, P, m)$, where:

1. $\mathcal{S}$ is the set of states of the Markov process, $\mathcal{S}$ is assumed finite unless otherwise mentioned.
2. $\mathcal{A}$ is the set of actions of P1 in each state, $\mathcal{A} = \{1, \ldots, A\}$.
3. $\mathcal{B}$ is the set of actions of P2 in each state, $\mathcal{B} = \{1, \ldots, B\}$.
4. $P$ is the state transition kernel. $P : \mathcal{S} \times \mathcal{A} \times \mathcal{B} \times \mathcal{S} \to [0, 1]$ such that $P(s_2|s_1, a_1, b_1)$ is the probability that the next state is $s_2$ given that current state is $s_1$, P1 plays $a_1$ and P2 plays $b_1$.
5. $m : \mathcal{S} \times \mathcal{A} \times \mathcal{B} \to \mathbb{R}^d$ is P1's vector reward function. $m(s, a, b)$ is the reward obtained when P1 plays $a$, P2 plays $b$ and the current state is $s$.

At each time epoch $n$, both players observe the current state $s_n$, and then P1 and P2 choose actions $a_n$ and $b_n$, respectively. As a result P1 receives a reward of $m_n = m(s_n, a_n, b_n)$ and the next state is determined according to the transition probability $P(\cdot|s_n, a_n, b_n)$. A strategy $\sigma_1 \in \Sigma_1$ for P1 is a mapping from all possible histories to the mixed action $\Delta^{\mathcal{A}}$, where $\Delta^{\mathcal{A}}$ is the set of all probability measures over $\mathcal{A}$. Similarly, a strategy $\sigma_2 \in \Sigma_2$ for P2 is a mapping from all possible histories to the mixed action $\Delta^{\mathcal{B}}$. For simplicity and without loss of generality we restrict attention to *behavioral* strategies where the mixing mechanisms at different stages are independent. A strategy of either player is called *stationary* if the mixed action in time $n$ depends only on the state $s_n$. The set of stationary strategies of P1 (resp. P2) is denoted by $F$ (resp. $G$). Let $P^s_{\sigma_1,\sigma_2}$ denote the probability measure induced on the sequence of states and actions by the strategy pair $\sigma_1$ and $\sigma_2$ and initial state $s$, and let $E^s_{\sigma_1,\sigma_2}$ denote the corresponding expectation operator. We are primarily interested in the average vector reward:

$$\hat{m}_n = \frac{1}{n}\sum_{l=1}^{n} m_l \,. \tag{2.1}$$

We shall also consider a related scalar-valued stochastic game that will be specified in the sequel. We denote the corresponding reward vector at time $n$ by $r_n = r(s_n, a_n, b_n) \in \mathbb{R}$. This reward may be stochastic, in which case $r(s_n, a_n, b_n)$ denotes its mean and the reward is assumed to have a finite second moment. In a similar manner to (2.1) the average vector valued reward is defined as $\hat{r}_n = \frac{1}{n}\sum_{l=1}^{n} r_l$. Given the vector-valued stochastic game and a vector $u \in \mathbb{R}^k$, we define the projected game $\Gamma_s(u)$ as the zero-sum stochastic game with (scalar) reward given by $r_n = m_n \cdot u$ and initial state $s$. $\Gamma_s(u)$ has a value $\mathrm{v}\Gamma_s(u)$ if

$$\begin{aligned} \mathrm{v}\Gamma_s(u) \quad &- \quad \sup_{\sigma_1}\inf_{\sigma_2}\liminf_{n\to\infty} E^s_{\sigma_1\sigma_2}(\hat{m}_n \cdot u) \\ &- \quad \inf_{\sigma_2}\sup_{\sigma_1}\limsup_{n\to\infty} E^s_{\sigma_1\sigma_2}(\hat{m}_n \cdot u)\,. \end{aligned} \tag{2.2}$$

As is well known ([MN81]) the value exists for any finite stochastic game.

We now recall the following definition that was first introduced in [Bla56], and later extended to stochastic games in [SS93]. Let $d(\cdot,\cdot)$ denote the Euclidean norm in $\mathbb{R}^k$. Recall that given a set of probability measures $\{P_\alpha\}_{\alpha\in\Upsilon}$ defined on a common measurable space, and a series $X_n$ of random variables on that space, we say that $X_n \to 0$ almost surely at a uniform rate over $\Upsilon$ if $\forall \epsilon > 0$

$$\lim_{N\to\infty}\sup_{\alpha\in\Upsilon} P_\alpha\left(\sup_{n\geq N}|X_n| > \epsilon\right) = 0\,.$$

**Definition 2.1** *A set $B \subseteq \mathbb{R}^k$ is* approachable *from state $s$ by P1 if there exists a $B$-*approaching *strategy $\sigma_1^*$ such that*

$$d(\hat{m}_n, B) \to 0 \quad P^s_{\sigma_1^*,\sigma_2}\text{-a.s. for every } \sigma_2 \in \Sigma_2$$

*at a uniform rate over* $\sigma_2 \in \Sigma_2$ *(i.e. over* $\{P_{\sigma_1^* \sigma_2}\}_{\sigma_2 \in \Sigma_2}$*).* $B$ *is* excludable *from* $s$ *by P2 if there exists a* $B$-excluding *strategy* $\sigma_2^* \in \Sigma_2$ *such that for some* $\delta > 0$

$$\liminf_{n \to \infty} d(\hat{m}_n, B) > \delta \quad P^s_{\sigma_1, \sigma_2^*}\text{-a.s. for every } \sigma_1 \in \Sigma_1$$

*at a uniform rate over* $\sigma_1 \in \Sigma_1$ *(i.e. over* $\{P^s_{\sigma_1, \sigma_2^*}\}_{\sigma_1 \in \Sigma_1}$*).*

# 3 Sufficient Conditions for Approachability

In this section we present an abstract approachability theorem. We later specialize this theorem to the case of a single state reachable structure. The abstract theorem requires no assumptions on the game dynamics. The renewal time of [SS93] is replaced with a general stopping time and the approaching strategy is based on increasing times between the policy switches. The theorem relies on the observation that if all the expansions of a set are approachable then so is the set itself. Let $\tau$ denote a positive finite (almost surely) stopping time, a strategy $\sigma_1 \in \Sigma_1$ is called *stable* with respect to $\tau$ and an initial state $s$ if $\exists M$ such that:

$$E^s_{\sigma_1 \sigma_2}(\tau^2) < M\,, \tag{3.1}$$

for any strategy $\sigma_2$ of P2 ($E^s_{\sigma_1, \sigma_2}$ is the expected value when starting from state $s$, with strategies $\sigma_1$ and $\sigma_2$). A set of strategies $\Psi_1 \subseteq \Sigma_1$ and corresponding stopping times $\tau(\sigma_1)$ is called *uniformly stable* with respect to a specific state $s$ and stopping times $\tau(\sigma)$ if every strategy $\sigma$ in $\Psi_1$ is stable with respect to $\tau(\sigma)$, with the same constant $M$ in (3.1).

Let

$$\phi^\tau_s(\sigma_1, \sigma_2) = \frac{E^s_{\sigma_1, \sigma_2}(\sum_{n=0}^{\tau-1} m_n)}{E^s_{\sigma_1, \sigma_2}(\tau)} \tag{3.2}$$

denote the average payoff until the stopping time from state $s$. Note that $\phi$ is well defined if either $\sigma_1$ or $\sigma_2$ is stable with respect to $s$ and $\tau$. Let $B$ be a closed set in $\mathbb{R}^d$, for any $x \notin B$ we denote by $C_x$ a closest point in $B$ to $x$. Let $H_x$ be the hyperplane through $C_x$ that is perpendicular to the line $C_x - x$ and $u_x$ be the unit vector in direction $C_x - x$.

We now provide a key lemma, the proof of this lemma is outlined, as the details are straightforward to verify. The lemma was stated for matrix games in [Spi99]. It is actually applicable in other situations, including Semi-Markov Decision Processes ([LS94]).

**Lemma 3.1** *Let* $B \subseteq \mathbb{R}^d$ *be set, and suppose that* $\forall \epsilon > 0$ *the* $\epsilon$ *expansion of* $B$*,* $B^\epsilon$*, is approachable. Then* $B$ *is approachable.*

**Proof:** (outline) The idea is to construct a strategy that approaches $B$ by approaching $B^\epsilon$ with $\epsilon \to 0$. Fix $\epsilon > 0$, then there exists a strategy such that $\forall n > N(\epsilon)$, with probability $1 - p_1$,

$$d(\hat{m}_n, B^\epsilon) \leq f_n(\epsilon)\,, \tag{3.3}$$

and $f_n(\epsilon) \to 0$. Fix a sequence of numbers $\beta_k \uparrow \frac{1}{2}$, $\beta_k > 0$. We propose a strategy and show that it approaches $B$ itself.

1. Play a strategy $(\sigma_1)$ that approaches $B^\epsilon$. Then $\exists N_1$ such that with probability $1 - p_1$ for $n > N_1$:

   $$d(\hat{m}_n, B^\epsilon) \leq \beta_1 \epsilon\,,$$

   and consequently by playing $\sigma_1$ for $n > N_1$

   $$d(\hat{m}_n, B) \leq (1 + \beta_1)\epsilon\,. \tag{3.4}$$

   $\sigma_1$ is played for $N_1$ steps.

2. Prepare to reduce $\epsilon$: Suppose that $\sigma_2$ is a $B^{\epsilon/2}$ approaching strategy. There exists a time $N'$ such that by playing $\sigma_2$ with probability $p_2$

   $$d(\hat{m}_n, B) \leq (1 + \beta_2)\epsilon/2\,. \tag{3.5}$$

   Assume that the reward is bounded by $C$ (or generally that the second moment is bounded.) Let the strategy $\sigma_1$ be played additional $2N'C/\epsilon$ epochs. Obviously, during this time (3.4) is satisfied.

3. Change to strategy $\sigma_2$: Play $\sigma_2$ for $N'$ time epochs. During this time we have that:

   $$d(\hat{m}_n, B) \leq \frac{N_1 + 2N'C/\epsilon}{n}(1 + \beta_1)\epsilon + \frac{n - (N_1 + 2N'C/\epsilon)}{n}C \tag{3.6}$$

   $$\leq 1.5\epsilon + 1/2\epsilon = 2\epsilon\,. \tag{3.7}$$

   Thus, $d(\hat{m}_n, B) \leq 2\epsilon$ during this time.

4. Complete the approaching of $B^{\epsilon/2}$: Play $\sigma_2$ for a long enough period such that (with probability (1-$p_2$)):

   $$d(\hat{m}_n, B) \leq \alpha(1 + \beta_1)\epsilon + (1 - \alpha)(1 + \beta_2)\epsilon/2\,,$$

   where $0 < \alpha < 1$ is determined by the length of this stage. Finally since $\beta_2 > \beta_1$ by setting $\alpha$ small enough there exists a time $N_2$ such that:

   $$d(\hat{m}_n, B) \leq (1 + \beta_2)\epsilon/2\,. \tag{3.8}$$

The whole process can be repeated with $\epsilon/4$ replacing $\epsilon/2$ and $\epsilon/2$ replacing $\epsilon$, and similarly $\beta_2$ replace $\beta_1$ and $\beta_3$ replace $\beta_2$. In the interval between $N_k$ and $N_{k+1}$ we get that

$$d(\hat{m}_n, B) \leq (1 + \beta_k)\epsilon 2^{-k+1} < \epsilon 2^{-k+2}\,,$$

with probability $(1-p_k)$. Now taking $p_k$ such that $\sum_{k=1}^{\infty} p_k < \infty$ and using the Borel-Cantelli Lemma concludes the proof. **Q.E.D.**

The theorem below is Blackwell's theorem for stochastic games, and is an abstract generalization of Theorem 3.1 of [SS93]. Note that additional stability or recurrence conditions are not required since they are replaced by the stability property of the stopping times $\tau$

**Theorem 3.2** *Assume that $\forall \delta > 0$, $\forall s \in \mathcal{S}$ and $\forall x \notin B^\delta$ there exist a finite stopping times $\tau(s,x,\delta)$ and a strategy $\sigma_1(s,x,\delta) \in \Sigma_1$ such that:*

$$(\phi_s^\tau(\sigma_1(x),\sigma_2) - C_x) \cdot u_x \geq -\delta \quad \forall \sigma_2 \in \Sigma_2 \,. \tag{3.9}$$

*Furthermore, assume that $\forall \delta$ the set $\{\sigma_1(s,x,\delta) : x \notin B^\delta, s \in \mathcal{S}\}$ is uniformly stable with respect to the corresponding stopping times. Then $B$ is approachable by P1.*

**Proof:** Observe that by Lemma 3.1 it suffices to show that for every $\epsilon > 0$, the set $B^\epsilon$ is approachable. The rest of the proof is very similar to the proof of Theorem 3.1 in [SS93], the main point is the observation that if (3.9) holds for a certain $\epsilon$ then a similar separation condition also holds for the closest point to $x$ in $B^\epsilon$, $C_x - \epsilon u_x$ (the minus sign is due to the definition of $u_x$ as a vector into $B$). Using this observation, (3.9) becomes (adding $\epsilon$ to both sides):

$$(\phi_s^\tau(\sigma_1(x),\sigma_2) - (C_x - \epsilon u_x)) \cdot u_x \geq -\delta + \epsilon \,. \tag{3.10}$$

By choosing $\delta = \epsilon$ we get the standard separation condition for $B^\epsilon$. That is (for $x \in \mathbb{R}^d \setminus B^\epsilon$):

$$(\phi_s^\tau(\sigma_1(x),\sigma_2) - C_x^\epsilon) \cdot u_x \geq 0 \,, \tag{3.11}$$

where $C_x^\epsilon$ is the closest point to $x$ in $B^\epsilon$. We can now give a similar proof to that of [SS93, Theorem 3.1] using martingale calculations. We provide the relations to the equations of [SS93, Theorem 3.1]. Fix $\epsilon > 0$, let $T(k)$ be the k-th time a stopping time occurred, by definition $T(0) = 0$. Let $d_t^\epsilon$ denote the distance of the average reward at time $t$ from $B^\epsilon$, $d_t^\epsilon = d(\hat{m}_t, B^\epsilon)$. Let $\tau_{k+1}$ be the k-th stopping time length, i.e. $\tau_{k+1} = T(k+1) - T(k)$. Let $v_{k+1}$ be the reward of the k-th strategy interval, that is

$$v_{k+1} \stackrel{\Delta}{=} \sum_{n=T(k)}^{T(k+1)-1} m_n = T(k+1)\hat{m}_{T(k+1)} - T(k)\hat{m}_{T(k)} \,. \tag{3.12}$$

Since the reward is assumed bounded we clearly have that

$$\begin{aligned} E\left(\tau_{k+1}^2 | \mathcal{F}_{T(k)}\right) &\leq M \\ E\left(v_{k+1}^2 | \mathcal{F}_{T(k)}\right) &\leq R \,, \end{aligned} \tag{3.13}$$

where $\mathcal{F}_t$ is the sigma algebra generated by the process over time epochs 1 to $t$. The main idea is to establish that

$$E\left(T(k+1)^2 (d_{T(k+1)}^\epsilon)^2 | \mathcal{F}_{T(k)}\right) \leq T(k)^2 (d_{T(k)}^\epsilon)^2 + W_k^2 \,, \tag{3.14}$$

where $d_t^\epsilon$ is the distance of $\hat{m}_t$ from the $\epsilon$ expansion of $B$, and $W_k$ satisfies $EW_k^2 \leq Q$ for some fixed $Q$. The condition in (3.11) is equivalent to requiring that: on $\{d_{T(k)} > \epsilon\}$

$$\begin{pmatrix} Ev_{k+1}|\mathcal{F}_{T(k)} \\ E\tau_{k+1}|\mathcal{F}_{T(k)} \end{pmatrix} - (C_{T(k)} - \epsilon u) \cdot u_{\hat{m}_{T(k)}} \geq 0 \tag{3.15}$$

Rearranging terms, as in [SS93, equation 16] we get that:

$$E\left(v_{k+1} - \tau_{k+1} C^\epsilon_{T(k+1)}\right) \cdot \left(C^\epsilon_{T(k)} - \hat{m}_{T(k)}\right) \geq 0\,, \tag{3.16}$$

where $C_t^\epsilon$ is the closest point to $\hat{m}_t$ in the $\epsilon$ expansion of $B$. Now we can follow equation (4.17) in [SS93] to find out that (3.14) holds. The rest of the proof follows [SS93] closely. The main steps there are to use a martingale convergence lemma to show that $\frac{1}{k} T(k) d^\epsilon_{T(k)} \to 0$, this implies that $d^\epsilon_{T(k)} \to 0$ and then to show that in between the stopping times $T(k)$, the distance can not increase too much. The result for $B$ itself follows. **Q.E.D.**

**Remark 1:** The above theorem holds without the finiteness assumption of $\mathcal{S}$. In fact, the state space may be arbitrary.

We now specialize this abstract theorem for the case of finite state space with a single reachable state structure. Before stating the theorem let us define a P1-0-reachable game.

**Definition 3.1** *A game is called* P1-0-reachable *if* $\forall s \in \mathcal{S}$ *there exists a strategy* $\sigma_1(s)$ *for P1 such that the renewal time from state* $s$ *to state 0,* $\tau$ *satisfies that* $\forall \sigma_2 \in \Sigma_2$

$$E^s_{\sigma_1(s),\sigma_2} \tau^2 < M$$

The above property may be difficult to verify in general. For finite games it is straightforward to check by inspecting the transition structure. This is the case since if P1 can ensure (almost surely) a return to state 0 from any state then the return is "fast" enough and the game is P1-0-reachable as the following lemma asserts.

**Lemma 3.3** *Assume the game is finite and suppose that P1 can reach state* 0 *with probability 1. Then the game is P1-0-reachable.*

**Proof:** Construct the following stochastic shortest path game. Let the states and actions of both players be the same. Let all the transitions are the same as the original game but the transition from state 0 which is now an absorbing state, and set all the rewards to +1 but the reward in state 0 which is set to 0. Since state 0 is reachable with probability 1, the assumptions of [Pat97, Proposition 3.2] apply. So that there exists a stationary optimal strategy $f_1^*$ for P1 in the stochastic shortest path game (P1 is a minimizer in this game). This implies that when $f_1^*$ is

played by P1 the expected time to reach state 0 is bounded (the number of states is finite). This implies that for the MDP induced by $f_1^*$ we have that for any strategy $\sigma_2 \in \Sigma_2$:

$$E_{f_1^* \sigma_2} \tau \leq M \,, \tag{3.17}$$

But it was established in [Bor88, page 74] that (3.17) implies that $\sup_{\sigma_2 \in \Sigma_2} E_{\sigma_2} \tau^2 < \infty$. **Q.E.D.**

We can now formulate an approachability theorem for arbitrary sets and P1-0-reachable games. This theorem is also a consequence of [Mil00, Theorem 2.1.1].

**Corollary 3.4** *Assume that the game is P1-0-reachable. Let $B \subset \mathbb{R}^d$ be the target set. Suppose that $\forall x \notin B$*

$$\Gamma_0(u_x) \geq u_x \cdot C_x \quad . \tag{3.18}$$

*Then $B$ is approachable by P1.*

**Proof:** We apply Theorem 3.2 by defining $\tau$ and $\sigma$. Fix $\delta > 0$, let $\sigma(x)$ be concatenated by playing the policy that returns to the reference state, until the return to reference state and then play $\sigma_1(x)$ for a long enough period. Let the stopping $\tau(x, s, \delta)$ time be defined as:

$$\tau(x, s, \delta) \stackrel{\Delta}{=} \sup_{\sigma_2 \in \Sigma_2} \min_t E \left\{ E_{\sigma(x)\sigma_2} \hat{m}_t \cdot u_x \geq -\delta \right\} . \tag{3.19}$$

We claim that $\tau(x, s, \delta)$ is finite, this is so because the expected renewal time of the return to state 0 is bounded by $(1 + M)$ ($M$ is from Definition 3.1), and the expected effect of the first part of $\sigma_1$ is bounded by $K(1 + M)$ ($K$ is some number that bounds the expected reward norm per cycle). The conditions of Theorem 3.2 are naturally satisfied by the definition of $\tau(x, s, \delta)$. Moreover, $\tau(x, s, \delta)$ is uniform for a given $\delta$ since that for finite games the convergence rate of the optimal policy to the value in an $n$ stage game converges uniformly (in the game parameters) to the expected value for the average cost game (e.g. [Mil00, Theorem 1.8.2]). **Q.E.D.**

# 4 Convex Sets

In order to establish a characterization of approachable convex sets, similarly to [Bla56], we have to let both players be able to reach a reference state. The basic assumption is that there is some reference state to which both players can return in bounded time. Similarly to Definition 3.1 we define a P2-0-reachable game.

**Definition 4.1** *A game is called* P2-0-reachable *if $\forall s \in \mathcal{S}$ there exists a strategy $\sigma_2(s)$ for P2 such that the renewal time $\tau$ from state $s$ to state 0, satisfies that $\forall \sigma_1 \in \Sigma_1$*

$$E^s_{\sigma_1, \sigma_2(s)} \tau^2 < M$$

Obviously an analogue result to Lemma 3.3 can be easily derived. We can now define in an analogue manner to [SS93] the sets $M(F,g)$. As the expectation from different states may differ, we refer to the expectation starting from the reference state 0. Define for every stationary strategy $g$ the set $M(F,g) \stackrel{\Delta}{=} \mathrm{co}(\{\lim_{n\to\infty} E^0_{f,g}\hat{m}_n, f \in F\})$, co stands for the closure of the convex hull. The following theorem characterizes convex approachable sets:

**Theorem 4.1** *Suppose that a finite game is P1-0-reachable and P2-0-reachable. Let $B$ be a closed convex set in $\mathbb{R}^k$.*

*(i) $B$ is approachable if and only if either one of the following equivalent conditions are satisfied:*

*a. $M(F,g) \cap B \neq \emptyset$ for every stationary strategy $g \in G$.*

*b. $\mathrm{v}\Gamma_0(u) \geq \min_{y\in B} y \cdot u$ for every $u \in U(B)$. $U(B) \subset \mathbb{R}^k$ is the set of all unit vectors $u$ in $\mathbb{R}^k$ such that $\exists x \notin B$ for which $C_x - x$ is in the $u$ direction.*

*(ii) If $B$ is not approachable then it is excludable by P2.*

**Proof:** Since any projected game $\Gamma(u)$ is both P1-0-reachable and P2-0-reachable, the value of the average reward game does depend on the initial state. Using [FV96, Theorem 5.3] we know that $\forall\epsilon > 0$ there is an $\epsilon$ optimal stationary strategy. First we establish the equivalence of [a] and [b]. Assume that $\forall g \in G$, $M(F,g) \cap B \neq \emptyset$, and let $g^\epsilon$ be an $\epsilon$ optimal stationary strategy at the some direction $u$. We have that:

$$\mathrm{v}\Gamma_0(u) \geq \sup_{f\in F} m(f,g^\epsilon)\cdot u + \epsilon, \tag{4.1}$$

where the supremum is over stationary strategies since for a stationary strategy of P2, P1 has an optimal stationary strategy. But $m(f,g^\epsilon)\cdot u \geq \min_{b\in B} b\cdot u$ so that $\mathrm{v}\Gamma_0(u) \geq \min_{b\in B} b\cdot u + \epsilon$. But $\epsilon$ is arbitrary, thus [b] is satisfied. Assume conversely that $M(F,g') \cap B = \emptyset$ for some $g'$ and that [b] is satisfied. Since both sets are closed and convex they are strongly separated in some direction $u'$, i.e.

$$\max_{x\in M(F,g')} x\cdot u' < \min_{b\in B} b\cdot u' - \epsilon,$$

But $F$ contains an $\epsilon/2$ optimal strategy in the $u'$ direction, so that $\max_{x\in M(F,g')} x\cdot u' > \mathrm{v}\Gamma_0(u') - \epsilon/2$, so we have that $\mathrm{v}\Gamma_0(u') < \min_{b\in B} b\cdot u' - \epsilon/2$, contradicting [b].

If [b] holds then the conditions for Corollary 3.4 apply, so that $B$ is approachable.

We now prove the second part of the theorem. If $\exists g$ stationary such that $M(F,g) \cap B = \emptyset$ then there is a direction $u$ such that there is a hyperplane that is supported by $u$ and that separates $B$ from $M(F,g)$. Let the projected distance between $B$ and $M(F,g)$ be $\epsilon$, that is $\inf_{b\in B} b\cdot u - \sup_{x\in M(F,g)} x\cdot u \geq \epsilon$. For the game $\Gamma_0(u)$ there is an $\epsilon/2$ optimal stationary strategy

$f \in F$, this implies that $v\Gamma_0(u) \leq \sup_{x \in M(F,g)} x \cdot u + \epsilon/2$. We can now apply Corollary 3.4 for P2 and the half space

$$\{x \in \mathbb{R}^d : x \cdot u \leq \sup_{x \in M(F,g)} x \cdot u + \epsilon/2\}.$$

This set is approachable by P2 since (3.18) clearly holds and therefore $B$ is excludable. **Q.E.D.**

From the proof above it stands out that the sufficient condition in the first part of the theorem holds if the game is merely P1-0-reachable, and the second part holds if the game is merely P2-0-reachable. This variant of the approachability theory might be useful, as demonstrated in [MS00]. It should also be noted that the actual assumption used is that the value for each projected game is independent of the initial state. Using the uniform convergence of the $n$ stage value to the value of the average reward game a generalized theorem can be proved.

A disadvantage of the approachability theorems presented is that convergence rate is not supplied. The reason is that actually the $\epsilon$ expansion of a set is approached rather than the set itself. Obtaining rate results, as in [SS93], requires that there will be a definite relation between $\delta$ and the statistics of the renewal time. In [SS93], the case of a single recurrent reference state is studied. It is required there that for $\delta = 0$ there is some finite $M$, that bounds the expected square return time to the reference state.

# 5 Infinite State Space

The case of infinite state is not as well structured as the finite case, as the uniformity and existence of value are not evident anymore. We have to rely on explicit assumptions regarding the stability and existence of limit for the expected reward of two stationary strategies.

The sufficient condition of Theorem 3.2 applies to infinite state space as well, as mentioned in Remark 1. A similar corollary to Corollary 3.4 can still can be stated for infinite state space if we impose that for every relevant $x \notin B$ the average reward of the strategy $\sigma_1(x)$ becomes high enough in that direction, uniformly with respect to the directions and the opponent's strategy. That is, we require that $\forall x \notin B$ there exists $\sigma_1(x)$ such that

$$E_{\sigma_1\sigma_2}(\hat{m}_n - C_x) \cdot u_x \geq 0 \quad \forall \sigma_2 \in \Sigma_2$$

and that $\{(\hat{m}_n - C_x) \cdot u_x\}^-$ converges [1] uniformly in $\sigma_2$ and $u$. We can then apply the same proof and the corollary still holds. Note that this does not depend on the existence of a value.

Before proceeding to characterization of convex sets some assumptions are to be taken. The reason is that we need both players to have a stable $\epsilon$ optimal stationary strategy in every direction. The basic assumption is that the value in each direction exists in general strategies.

[1] The operator $\{x\}^- \triangleq \min(0, x)$

**Assumption 1** *$\forall u \in U$ the associated projected game $\Gamma(u)$ the value exists and equals $v\Gamma(u)$.*

We require the existence of limit average for stationary strategies starting from the reference state.

**Assumption 2** *$\forall f \in F, g \in G$ the limit $\lim_{n\to\infty} E^0_{fg}\hat{m}_n$ exists.*

See [DY75] for conditions that guarantee this assumption. We now make additional assumption for the existence of $\epsilon$ optimal stationary strategies

**Assumption 3** *Let Assumption 1 hold. Then $\forall \epsilon > 0$ and $\forall u \in U$*

*i. There is an $\epsilon$ optimal stationary for P1, $f(u)$ such that*

$$\mathrm{v}\Gamma_0(u) - \epsilon \leq \inf_{\sigma_2} \liminf_{n\to\infty} E^0_{f(u)\sigma_2}(\hat{m}_n \cdot u)\,.$$

*ii. There is an $\epsilon$ optimal stationary for P2, $g(u)$ such that*

$$\mathrm{v}\Gamma_0(u) + \epsilon \geq \sup_{\sigma_1} \limsup_{n\to\infty} E^0_{\sigma_1 g(u)}(\hat{m}_n \cdot u)\,.$$

There are conditions under which the assumption is satisfied, these conditions are related to the uniform integrability of the reward, or uniform Liapunov functions, see (in MDP context) e.g. [Alt99, chapter 11].

We are now equipped with the assumptions that ensure that an $\epsilon$ optimal stationary strategy exist. We now need assumptions that are related to the stability of such strategies. As in the finite case the basic assumption is that the game is P1-0-reachable and P2-0-reachable. We actually require a weaker assumption, as it may be easily verified that if a game is P1-0-reachable and P2-0-reachable then the following assumption holds:

**Assumption 4** *There exists a state $0 \in \mathcal{S}$ such that*

*i. $\forall \delta > 0$ and for every stationary strategy $f$ of P1 there exists a strategy $\sigma_f$ for P1 such that for large $n$:*

$$||E^s_{\sigma_f g}\hat{m}_n - E^0_{fg}\hat{m}_n|| \leq \delta\,. \tag{5.1}$$

*Furthermore, when playing $\sigma_f$ the rate of convergence of*

$$\left\{||\hat{m}_n - E^0_{fg}\hat{m}_n|| - \delta\right\}^+ \tag{5.2}$$

*is [2] uniform in $g \in G$, $P_{\sigma_f g}$ almost surely.*

[2]The operator $\{x\}^+ \triangleq \max\{0, x\}$

*ii.* $\forall \delta > 0$ *and for every stationary strategy* $g$ *of P2 there exists a strategy* $\sigma_g$ *for P2 such that for large* $n$*:*

$$||E^s_{f\sigma_g}\hat{m}_n - E^0_{fg}\hat{m}_n|| \leq \delta\,. \tag{5.3}$$

*Furthermore, when playing* $\sigma_g$ *the rate of convergence of*

$$\left\{||\hat{m}_n - E^0_{fg}\hat{m}_n|| - \delta\right\}^+ \tag{5.4}$$

*is uniform in* $f \in f$*,* $P_{f\sigma_g}$ *almost surely.*

We can now define in an analogue manner to [SS93] the sets $M(F,g)$. As the expectation from different states is no longer the same we refer to the expectation starting from the reference state 0. If Assumption 2 holds, we then define for every stationary strategy $g$ the set $M(F,g) \stackrel{\Delta}{=} \text{co}(\{\lim_{n\to\infty} E^0_{f,g}\hat{m}_n, f \in F\})$, co stands for the closure of the convex hull. The following theorem characterizes convex approachable sets:

**Theorem 5.1** *Let Assumptions 1-4 hold. Let* $B$ *be a closed convex set in* $\mathbb{R}^k$*.*

*(i)* $B$ *is approachable if and only if either one of the following equivalent conditions are satisfied:*

*a.* $M(F,g) \cap B \neq \emptyset$ *for every stationary strategy* $g \in G$*.*

*b.* $\text{v}\Gamma_0(u) \geq \min_{y\in B} y \cdot u$ *for every* $u \in U(B)$*.* $U(B) \subset \mathbb{R}^k$ *is the set of all unit vectors* $u$ *in* $\mathbb{R}^k$ *such that* $\exists x \notin B$ *for which* $C_x - x$ *is in the* $u$ *direction.*

*(ii) If* $B$ *is not approachable then it is excludable by P2.*

**Proof:** Assumptions 3 and 4 implies that $\forall \delta > 0$ and for every direction $u$ there exist an $\epsilon + \delta$ optimal strategy that satisfies (5.2) uniformly. The equivalence of [a] and [b] is established similarly to the finite state space case, as in Theorem 4.1.

We need to show if [b] holds then the conditions for Theorem 3.2 apply. Fix $\delta > 0$, it obviously suffices to prove the conditions of Theorem 3.2 for every direction $u$ rather than for every $x \notin B$. The stopping time $\tau$ is defined as:

$$\tau(u,\delta) \stackrel{\Delta}{=} \sup_{\sigma_2\in\Sigma_2} \min_t E\left\{\frac{1}{t}\sum_{n=1}^{t} m_n \cdot u \geq \text{v}\Gamma_0(u) - \delta\right\}. \tag{5.5}$$

$\tau$ is a stopping time and it remains to explain which strategy $\sigma_1(u)$ is chosen at every epoch the stopping time $\tau$ is activated and to prove that

$$E_{\sigma_1(u),\sigma_2}\tau^2 < M\,, \tag{5.6}$$

for all $\sigma_2$, uniformly in $u \in U$ (the uniformity in $u$ is required so that the conditions of Theorem 3.2 are satisfied) . We now sketch the construction of $\sigma_1(u)$ and the proof that (5.6) holds. The

full details are omitted as the proof is straight-forward and does not present an optimal way to construct the best strategy.

For every direction $u \in U(B)$ there exists a $\delta/3$ optimal stationary strategy, $f(u)$ that is a strategy that ensures that the $u$ projected expected average reward is at least $\mathrm{v}\Gamma_0(u) - \delta/3$. For every direction $u$ denote by $N_u$ the set of directions $u' \in U(B)$ such that by playing $f(u)$, the average reward satisfies for large enough $n$ that $||\hat{m}_n \cdot u' - \hat{m}_n \cdot u|| \leq \delta/3$. Since the inner product is continuous and the vector-valued expected reward $m$ is bounded it is straightforward to show that $N_u$ contains an open neighborhood around $u$. Since $U$ is compact (or at least its closure is) there exists a finite number of directions $U' = u_1, \ldots, u_K$ such that $U \subseteq \bigcup_{i=1}^{K} N_{u_i}$. P1's strategy at every time $\tau$ is to play $\sigma_{f(u')}$ where $\sigma_f$ is defined in Assumption 4 for some $u' \in U'$ for which $u \in N_{u'}$. Combining the above we have that by choosing $\sigma_{f(u)}$ the average reward satisfies $\hat{m}_n \cdot u - \mathrm{v}\Gamma(u) \geq -2\delta/3$ for large $n$.

It suffices to prove that $E_{g(u_i)\sigma_2}\tau^2 < M$ for every $u_i \in U'$ and $\sigma_2$. This result follows by noticing that there is a finite number of MDPs involved and by the uniformity in Assumption 4.

The second part of the theorem follows exactly as the finite case (Theorem 4.1). **Q.E.D.**

## 6 An Example

In this section we provide a simple example for a model which does not have a recurrent state under all stationary strategies, but still an approachability framework can be devised. Let $\mathcal{S} = \{z, x, y\}$. $\mathcal{A}(z) = \{1\}$, $\mathcal{A}(x) = \{1, 2\}$, $\mathcal{A}(y) = \{1, 2\}$. There is no opponent in this game, so that $\mathcal{B} = \{1\}$ for all states. The state transition is satisfies $p(s' = x|s = z) = p(s' = y|s = z) = 1/2$, that is the move from state $z$ to either state $x$ or $y$ is with probability $1/2$. The transition probabilities from states $x$ and $y$ are deterministic so that $p(s' = z|s = x, a = 1) = 1, p(s' = x|s = x, a = 2) = 1$ and similarly $p(s' = z|s = y, a = 1) = 1, p(s' = y|s = y, a = 2) = 1$. The reward is two dimensional vector so that $r(z, 1) = (0, 0), r(x, 1) = r(y, 1) = (-100, -100), r(x, 2) = (1, 0), r(y, 2) = (0, 1)$. In words, remaining in states $x$ or $y$ result in movement in X or Y directions, respectively. Picking action 1 in either states results in a return to state $z$ but the penalty is relatively high. This game does not have a recurrent state, as some deterministic strategies cause that either state $x$ or $y$ are absorbing. P1 can obviously ensure a fast return to state $z$ which is the reference state. Note then when only a single player is involved, the reachability should be satisfied only by that player. We therefore write $M(F) = M(F, g)$ since $g$ is unique.

Let the target set be the point $B = \{(1/2, 1/2)\}$. There is no stationary strategy that attain $B$. Indeed, if $f$ is stationary, it must be stochastic in both states (or else it will be absorbed in either states $x$ or $y$). Let the occupation frequencies of states $\{z, x, y\}$ be denoted by $\pi_z, \pi_x, \pi_y$, respectively. Let the frequency of action 1 in states $x$ and $y$ be denoted by $f_x$ and $f_y$, respectively.

We have that

$$\begin{aligned} \pi_z &= \pi_x f_x + \pi_y f_y \\ \pi_x &= \pi_x(1-f_x) + \pi_z/2 \\ \pi_z + \pi_x + \pi_y &= 1 \end{aligned} \tag{6.1}$$

So that:

$$\pi_z + \frac{\pi_z}{2f_y} + \frac{\pi_z}{2f_y} = 1\,.$$

That is $\pi_z = \frac{1}{1+\frac{1}{2f_x}+\frac{1}{2f_y}}$ and the average vector reward is:

$$\begin{aligned} \hat{m} &= (1,0)\pi_x(1-f_x) + (0,1)\pi_y(1-f_y) + (-100,-100)(\pi_x f_x + \pi_y f_y) \\ \hat{m} &= (1,0)\frac{1-f_y}{2f_y+1+\frac{f_x}{f_y}} + (0,1)\frac{1-f_y}{2f_x+1+\frac{f_y}{f_x}} + (-100,-100)\frac{1}{1+\frac{1}{2f_x}+\frac{1}{2f_y}} \end{aligned} \tag{6.2}$$

Thus we have that for $f_x = f_y = \epsilon$, with $\epsilon$ small enough $\hat{m}$ approaches $(\frac{1}{2},\frac{1}{2})$. As a result $B \cap M(F) = B$.

We now state an explicit approaching strategy. In the $k$-th iteration play action 2 for $\lfloor\sqrt{k}\rfloor$ epochs[3], then play action 1 and repeat the process. It can still be shown that this scheme approaches $B$. Indeed, let $Z_t$ denote the number of time epochs that state $x$ was in by time $t$. It suffices to show that $\frac{Z_t}{t} \to \frac{1}{2}$ almost surely. The amount of time spent in state z, and therefore the contribution of state $z$ goes to zero, as $k$ increases. Let $Y_k$ denote the time spent in state $x$ in the $k$th period. $Y_k$ is a random variable that satisfies:

$$Y_k = \begin{cases} \lfloor\sqrt{k}\rfloor & \text{with probability } \frac{1}{2} \\ 0 & \text{with probability } \frac{1}{2} \end{cases}$$

Let $N_k$ denote the time in which the $k$ renewal to state $z$ occurred. Denote by $X_k \triangleq Z_{N_k}$ the total time state $x$ was occupied until the $k$-th return. Obviously, $X_k = \sum_{i=1}^{k} Y_k$. Let $\tilde{Y}_k$ denote the centralized value of $Y_k$, that is $\tilde{Y}_k$ behaves like

$$\tilde{Y}_k = \begin{cases} \lfloor\sqrt{k}\rfloor/2 & \text{with probability } \frac{1}{2} \\ -\lfloor\sqrt{k}\rfloor/2 & \text{with probability } \frac{1}{2} \end{cases}$$

It suffices to show that the centralized sum $\tilde{X}_k = \sum_{i=1}^{k} \tilde{Y}_k$, satisfies that $\frac{\tilde{X}_k}{N_k} \to 0$.

$$E\tilde{X}_k^2 = \lfloor\sqrt{i}\rfloor^2/4 \le i/4\,,$$

and

$$N_k = \sum_{i=1}^{k}\lfloor\sqrt{i}\rfloor + k \ge \sum_{i=1}^{k}(\sqrt{i}-1) + k \ge \int_0^{k-1}\sqrt{x}dx = 2/3(k-1)^{3/2} \tag{6.3}$$

[3] $\lfloor x\rfloor$ denotes the largest integer that is lower than $x$

We can now apply [Dur91, Theorem 8.2]. Since

$$\sum_{i=1}^{\infty} \frac{E\tilde{X}_k^2}{N_k^2} \leq 1 + \sum_{i=2}^{\infty} \frac{i}{2/3(i-1)^{3/2}2} \leq C' + C\sum_{i=1}^{\infty} \frac{1}{i^2} < \infty, \tag{6.4}$$

we have that $\frac{\tilde{X}_k}{N_k} \to 0$ almost surely. We therefore have that at the returns to state $z$, $Z_t/t \to 1/2$. Since the excursion times between returns to state $z$ behave like $\sqrt{t}$ the result follows also for $t$ in between the renewal times.

It should be noted that in the direction of $u = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ the value of the game is $\frac{1}{\sqrt{2}}$ but no stationary strategy achieves it, so clearly condition C1 of [SS93] is not satisfied.

# 7 Conclusion

We provided an extension of approachability theory for stochastic games to the case of no a-priori recurrent state. This result is of particular interest for game with a single reachable state structure. The results were formulated also for games with infinite state space under the proper assumptions.

A natural and straightforward extension of our result is to Semi Markov Decision Processes. Approachability theory for Semi Markov decision processes under recurrence conditions was suggested by [LS94]. Other possible extensions include work with infinite actions spaces ([Leh98]), study of weak approachability ([Vie92]), and defining approachability per state (as in [Mil00]).

Finally, we showed that if $B^\epsilon$ is approachable $\forall \epsilon > 0$ then so is $B$. We did that by constructing a strategy that is $B$ approaching. The rate of convergence, however, was obviously non optimal. It might be of great interest to investigate further approaching strategies with better rates. It is known that the rate of approachability for the standard case is at worst $\Theta(t^{-1/3})$, it is an open question what is the guaranteed rate for a model of the style described in this paper. We conjecture that this scheme behaves as $\Theta(t^{-1/3})$ as well.

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